

STROCSS 2021: Strengthening the reporting of cohort, cross-sectional and case-control studies in surgery

Item no.	Item description	Reported on Page Number/Line Number	Reported on Section/Paragraph
TITLE			
1	Title		
	- The word cohort or cross-sectional or case-control is included*	Page1/Line 2	Title
	- Temporal design of study is stated (e.g. retrospective or prospective)	Page1/Line 2	Title
	- The focus of the research study is mentioned (e.g. population, setting, disease, exposure/intervention, outcome etc.)	Page1/Line 1-2	Title
ABSTRACT			
2a	Introduction – briefly describe:		
	- Background	Page 3/Line 67-69	Abstract/ Paragraph 1
	- Scientific rationale for this study	Page 3/Line 68-70	Abstract/ Paragraph 1
	- Aims and objectives	Page 3/Line 71-72	Abstract/ Paragraph 1
2b	Methods - briefly describe:		
	- Type of study design (e.g. cohort, case-control, cross-sectional etc.)	Page 3/Line 73	Abstract/ Paragraph 2
	- Other key elements of study design (e.g. retro-/prospective, single/multi-centred etc.)	Page 3/Line 73	Abstract/ Paragraph 2
	- Patient populations and/or groups, including control group, if applicable	Page 3/Line 74	Abstract/ Paragraph 2
	- Exposure/interventions (e.g. type, operators, recipients, timeframes etc.)	Page 3/Line 75	Abstract/ Paragraph 2
	- Outcome measures – state primary and secondary outcome(s)	Page 3/Line 76	Abstract/ Paragraph 2
2c	Results - briefly describe:		
	- Summary data with qualitative descriptions and statistical relevance, where appropriate	Page 3/Line 77-82	Abstract/Paragraph 3
2d	Conclusion - briefly describe:		
	- Key conclusions	Page 3/Line 83	Abstract/Paragraph 4
	- Implications for clinical practice	Page 3/Line 85	Abstract/Paragraph 4
	- Need for and direction of future research	Page 3/Line 85	Abstract/Paragraph 4

INTRODUCTION			
3	Introduction – comprehensively describe:		
	- Relevant background and scientific rationale for study with reference to key literature	Page 4/ L 90-124	Intro/ para 1-3
	- Research question and hypotheses, where appropriate	Page 5/ L125-L126	Intro/ para 4
	- Aims and objectives	Page 5/ L127-129	Intro/ para 4
METHODS			
4a	Registration		
	- In accordance with the Declaration of Helsinki [#] , state the research registration number and where it was registered, with a hyperlink to the registry entry (this can be obtained from ResearchRegistry.com, ClinicalTrials.gov, ISRCTN etc.)	Page 6/ Line 139	Methods/ para 1
	- All retrospective studies should be registered before submission; it should be stated that the research was retrospectively registered	Page 6/ Line 140	Methods/ para 1
4b	Ethical approval		
	- Reason(s) why ethical approval was needed	Page 6/ Line 138	Methods/ para 1
	- Name of body giving ethical approval and approval number	Page 6/ Line 140	Methods/ para 1
	- Where ethical approval wasn't necessary, reason(s) are provided	-	-
4c	Protocol		
	- Give details of protocol (<i>a priori</i> or otherwise) including how to access it (e.g. web address, protocol registration number etc.)	NA	NA
	- If published in a journal, cite and provide full reference	NA	NA
4d	Patient and public involvement in research		
	- Declare any patient and public involvement in research	NA	NA
	- State the stages of the research process where patients and the public were involved (e.g. patient recruitment, defining research outcomes, dissemination of results etc.) and describe the extent to which they were involved.	NA - retrospective	NA
5a	Study design		
	- State type of study design used (e.g. cohort, cross-sectional, case-control etc.)	Page 6/ L134	Methods/ para 1
	- Describe other key elements of study design (e.g. retro-/prospective, single/multi-centred etc.)	Page 6/ L134	Methods/ para 1

5b	Setting and timeframe of research – comprehensively describe:		
	- Geographical location	Page 6/ L137	Methods/ para 1
	- Nature of institution (e.g. primary/secondary/tertiary care setting, district general hospital/teaching hospital, public/private, low-resource setting etc.)	Page 6/ L134	Methods/ para 1
	- Dates (e.g. recruitment, exposure, follow-up, data collection etc.)	Page 6/ L135	Methods/ para 1
5c	Study groups		
	- Total number of participants	Page 9/ L 245	Results/ para 1
	- Number of groups	Page 6/ L 136-137	Methods/ para 1
	- Detail exposure/intervention allocated to each group	Page 6/ L 136-137	Methods/ para 1
	- Number of participants in each group	Page9/ L246-248	Results/para 1
5d	Subgroup analysis – comprehensively describe:		
	- Planned subgroup analyses	NA	NA
	- Methods used to examine subgroups and their interactions	NA	NA
6a	Participants – comprehensively describe:		
	- Inclusion and exclusion criteria with clear definitions	Page 6/ L 134/135	Methods/ para 1/ Figure 1
	- Sources of recruitment (e.g. physician referral, study website, social media, posters etc.)	NA retrospective	NA
	- Length, frequency and methods of follow-up (e.g. mail, telephone etc.)	Page 6/ L150	Methods/para 2
6b	Recruitment – comprehensively describe:		
	- Methods of recruitment to each patient group (e.g. all at once, in batches, continuously till desired sample size is reached etc.)	NA	NA
	- Any monetary incentivisation of patients for recruitment and retention should be declared; clarify the nature of any incentives provided	NA	NA
	- Nature of informed consent (e.g. written, verbal etc.)	Page 6/ L 141	Methods/ para 1
	- Period of recruitment	NA	NA
6c	Sample size – comprehensively describe:		
	- Analysis to determine optimal sample size for study accounting for population/effect size	NA	NA
	- Power calculations, where appropriate	NA	NA
	- Margin of error calculation	NA	NA

METHODS - INTERVENTION AND CONSIDERATIONS			
7a	Pre-intervention considerations – comprehensively describe:		
	- Preoperative patient optimisation (e.g. weight loss, smoking cessation, glycaemic control etc.)	NA	NA
	- Pre-intervention treatment (e.g. medication review, bowel preparation, correcting hypothermia/-volemia/-tension, mitigating bleeding risk, ICU care etc.)	NA	NA
7b	Intervention – comprehensively describe:		
	- Type of intervention and reasoning (e.g. pharmacological, surgical, physiotherapy, psychological etc.)	NA	NA
	- Aim of intervention (preventative/therapeutic)	NA	NA
	- Concurrent treatments (e.g. antibiotics, analgesia, antiemetics, VTE prophylaxis etc.)	NA	NA
	- Manufacturer and model details, where applicable	NA	NA
7c	Intra-intervention considerations – comprehensively describe:		
	- Details pertaining to administration of intervention (e.g. anaesthetic, positioning, location, preparation, equipment needed, devices, sutures, operative techniques, operative time etc.)	NA	NA
	- Details of pharmacological therapies used, including formulation, dosages, routes, and durations	NA	NA
	- Figures and other media are used to illustrate	NA	NA
7d	Operator details – comprehensively describe:		
	- Requirement for additional training	NA	NA
	- Learning curve for technique	NA	NA
	- Relevant training, specialisation and operator's experience (e.g. average number of the relevant procedures performed annually)	NA	NA
7e	Quality control – comprehensively describe:		
	- Measures taken to reduce inter-operator variability	NA	NA
	- Measures taken to ensure consistency in other aspects of intervention delivery	NA	NA
	- Measures taken to ensure quality in intervention delivery	NA	NA
7f	Post-intervention considerations – comprehensively describe:		
	- Post-operative instructions (e.g. avoid heavy lifting) and care	NA	NA
	- Follow-up measures	NA	NA
	- Future surveillance requirements (e.g. blood tests, imaging etc.)	NA	NA

8	Outcomes – comprehensively describe:		
	- Primary outcomes, including validation, where applicable	Page 6/ L 144 - 145	Methods / para 2
	- Secondary outcomes, where appropriate	Page 6/ L 146-148	Methods / para 2
	- Definition of outcomes	Page 8/ L193- 222	Methods/ para 8
	- If any validated outcome measurement tools are used, give full reference	NA	NA
	- Follow-up period for outcome assessment, divided by group	Page 6/L 145	Methods/ para 1
9	Statistics – comprehensively describe:		
	- Statistical tests and statistical package(s)/software used	Page 9/ L225-233	Methods/para 10
	- Confounders and their control, if known	Page 9/ L227	Methods/para 10
	- Analysis approach (e.g. intention to treat/per protocol)	NA	NA
	- Any sub-group analyses	NA	NA
	- Level of statistical significance	Page 9/ L227	Methods/para 10
RESULTS			
10a	Participants – comprehensively describe:		
	- Flow of participants (recruitment, non-participation, crossover and withdrawal, with reasons). Use figure to illustrate.	Page 10/ L245	Results/ para 1
	- Population demographics (e.g. age, gender, relevant socioeconomic features, prognostic features etc.)	Page 11/ L262-272	Results/ para 2
	- Any significant numerical differences should be highlighted	Page 11/ L262-272	Results/ para 2
10b	Participant comparison		
	- Include table comparing baseline characteristics of cohort groups	Page 11/ L262-272	Results/ para 2
	- Give differences, with statistical relevance	Page 11/ L262-272	Results/ para 2
	- Describe any group matching, with methods	Page 11/ L257	Results/ figure 2
10c	Intervention – comprehensively describe:		
	- Degree of novelty of intervention	Page 4/ L103	Intro/ para 1
	- Learning required for interventions	NA	NA
	- Any changes to interventions, with rationale and diagram, if appropriate	NA	NA

11a	Outcomes – comprehensively describe:		
	- Clinician-assessed and patient-reported outcomes for each group	Page 14/ L 294	Results/ Table 1
	- Relevant photographs and imaging are desirable	NA	NA
	- Any confounding factors and state which ones are adjusted	Page 11/ L257	Results/ figure 2
11b	Tolerance – comprehensively describe:		
	- Assessment of tolerability of exposure/intervention	Page 13/ L273-301	Results/ para 4
	- Cross-over with explanation	NA	NA
	- Loss to follow-up (fraction and percentage), with reasons	NA	NA
11c	Complications – comprehensively describe:		
	- Adverse events and classify according to Clavien-Dindo classification [†]	Page 9/ L 221	Methods/ para 9 (+table 3)
	- Timing of adverse events	Page 13/ L 284	Results/ para 3 (+table 3)
	- Mitigation for adverse events (e.g. blood transfusion, wound care, revision surgery etc.)	NA	NA
12	Key results – comprehensively describe:		
	- Key results with relevant raw data	Page 12/ L 273 - 294	Results/ para 4
	- Statistical analyses with significance	Page 12/ L 273 - 294	Results/ para 4
	- Include table showing research findings and statistical analyses with significance	Page 14/ L294, 295, 298	Results/ Table 1, 2, 3
DISCUSSION			
13	Discussion – comprehensively describe:		
	- Conclusions and rationale	Page 18/ L 331-372	Discussion/ para 3
	- Reference to relevant literature	Page 18/ L 309-327	Discussion/ para 1
	- Implications for clinical practice	Page 21/ L 386	Discussion/ para 8
	- Comparison to current gold standard of care	Page 21/ L 386	Discussion/ para 8
	- Relevant hypothesis generation	Page 21/ L 386	Discussion/ para 8

14	Strengths and limitations – comprehensively describe:		
	- Strengths of the study	Page 20/ L 384	Discussion/ para 7
	- Weaknesses and limitations of the study and potential impact on results and their interpretation	Page 20/ L 377	Discussion/ para 7
	- Assessment and management of bias	Page 20/ L 385	Discussion/ para 7
	- Deviations from protocol, with reasons	NA	NA
15	Relevance and implications – comprehensively describe:		
	- Relevance of findings and potential implications for clinical practice	Page 21/ L 386	Discussion/ para 8
	- Need for and direction of future research, with optimal study designs mentioned	Page 21/ L 386	Discussion/ para 8
CONCLUSION			
16	Conclusions		
	- Summarise key conclusions	Page 21/ L 386	Discussion/ para 8
	- Outline key directions for future research	Page 21/ L 388	Discussion/ para 8
DECLARATIONS			
17a	Conflicts of interest		
	- Conflicts of interest, if any, are described	NA	NA
17b	Funding		
	- Sources of funding (e.g. grant details), if any, are clearly stated	NA	NA
	- Role of funder	NA	NA
17c	Contributorship		
	- Acknowledge patient and public involvement in research; report the extent of involvement of each contributor	NA	NA

* STROCCS 2021 guidelines apply to cohort studies as well as other observational studies (e.g. cross-sectional, case-control etc.).

“Every research study involving human subjects must be registered in a publicly accessible database before recruitment of the first subject”.

† Dindo D, Demartines N, Clavien P-A. Classification of Surgical Complications. A New Proposal with Evaluation in a Cohort of 6336 Patients and Results of a Survey. Ann Surg. 2004; 240(2): 205-213.

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*As the checklist was provided upon initial submission, the page number/line number reported may be changed due to copyediting and may not be referable in the published version. In this case, the section/paragraph may be used as an alternative reference.